

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – Scenario-Based

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4-Scenario Based	Assessment:	Assessment Tool 1-Scenario Based
Weightage:	40% of CIA	Total Marks:	100
CO 4 Statement:	Analyze and apply point estimation techniques to estimate population parameters from sample data for effective statistical inference. (BL4)		
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Section / Batch:	CSE AI 2024	Date:	17-08-2026

Code of Conduct

I Y SRAVAN KUMAR (192472289) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ SRAVAN _____

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

CO4-AT1 - Scenario Based Questions

11.Hypothesis Testing Using Confidence Interval

Question

A university claims that at least 80% of its students are satisfied with online classes. A survey produces a 95% confidence interval of (74%, 79%) for the satisfaction proportion. What conclusion can be made about the university's claim?

Solution

Given:

Claimed satisfaction proportion:

$$p_0 = 80\% = 0.80$$

95% Confidence Interval:

$$CI = (74\%, 79\%)$$

Step 1: State the Hypotheses

$$H_0: p \geq 0.80$$

$$H_1: p < 0.80$$

Step 2: Compare the Confidence Interval

The claimed value is:

$$p_0 = 80\%$$

The 95% confidence interval is:

$$CI = (74\%, 79\%)$$

The entire confidence interval is below 80%.

Step 3: Decision

Since 80% is outside the 95% confidence interval, the sample evidence does not support the university's claim.

Conclusion

Therefore, the university's claim that at least 80% of students are satisfied with online classes is not supported by the survey data. The actual satisfaction proportion is likely to be below 80%.

12.Hypothesis Testing Using Confidence Interval

Question

A mobile application developer claims that users spend an average of 30 minutes per day on the application. A sample produces a 99% confidence interval of (31.5, 35.2) minutes. Using the confidence interval, determine whether the claim is supported.

Solution

Given:

Claimed mean:

$$\mu_0 = 30 \text{ minutes}$$

99% Confidence Interval:

$$CI = (31.5, 35.2) \text{ minutes}$$

Step 1: State the Hypotheses

$$H_0: \mu = 30$$

$$H_1: \mu \neq 30$$

Step 2: Compare the Confidence Interval

The claimed value is:

$$\mu_0 = 30 \text{ minutes}$$

The 99% confidence interval is:

$$CI = (31.5, 35.2) \text{ minutes}$$

Since 30 is less than 31.5, the claimed value of 30 minutes is not contained within the confidence interval.

Step 3: Decision

Since 30 minutes lies **outside the 99% confidence interval, the claim is not supported.**

Conclusion

Therefore, the developer's claim that users spend an average of 30 minutes per day is not supported by the sample data. The data suggests that the true average usage is greater than 30 minutes.

13. Hypothesis Testing Using p-value

Question

A credit-card company claims that its fraud detection system identifies 95% of fraudulent transactions. After testing the system on a sample, the calculated p-value is 0.02. At a 5% significance level, what conclusion should the data scientist make?

Solution

Given:

Claimed detection rate:

$$p_0 = 95\% = 0.95$$

p-value:

$$p = 0.02$$

Significance level:

$$\alpha = 0.05$$

Step 1: State the Hypotheses

$$H_0: p = 0.95$$

$$H_1: p \neq 0.95$$

Step 2: Compare p-value and Significance Level

$$\text{p-value} = 0.02$$

$$\alpha = 0.05$$

$$0.02 < 0.05$$

Step 3: Decision

Since $\text{p-value} < \alpha$, we reject H_0 .

Conclusion

There is statistically significant evidence that the fraud detection system's actual detection rate differs from the claimed 95% at the 5% significance level.

Note: The p-value alone does not tell whether the actual detection rate is higher or lower than 95%.

14.Hypothesis Testing Using p-value

Question

An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The hypothesis test produces a p-value of 0.03. At $\alpha = 0.05$, determine whether the new algorithm significantly improves conversion.

Solution

Given:

Existing conversion rate:

$$p_0 = 12\% = 0.12$$

Observed conversion rate:

$$\hat{p} = 15\% = 0.15$$

p-value:

$$p = 0.03$$

Significance level:

$$\alpha = 0.05$$

Step 1: State the Hypotheses

$$H_0: p = 0.12$$

$$H_1: p > 0.12$$

The alternative hypothesis is greater than 12% because the company wants to determine whether conversion has improved.

Step 2: Compare p-value and α

$$\text{p-value} = 0.03$$

$$\alpha = 0.05$$

$$0.03 < 0.05$$

Step 3: Decision

Since $\text{p-value} < \alpha$, we reject H_0 .

Conclusion

The new recommendation algorithm significantly improves the conversion rate at the 5% significance level.

15. Hypothesis Testing Using p-value

Question

A hospital introduces an AI-based patient scheduling system. The hospital wants to determine whether it reduces average waiting time. A hypothesis test produces a p-value of 0.08. Using a significance level of 5%, should the hospital conclude that the new system reduces waiting time?

Solution

Given:

p-value:

$$p = 0.08$$

Significance level:

$$\alpha = 0.05$$

Step 1: State the Hypotheses

H_0 : The new system does not reduce average waiting time.

H_1 : The new system reduces average waiting time.

Step 2: Compare p-value and α

p-value = 0.08

$\alpha = 0.05$

$0.08 > 0.05$

Step 3: Decision

Since p-value $> \alpha$, we fail to reject H_0 .

Conclusion

The hospital should not conclude that the AI-based scheduling system significantly reduces waiting time at the 5% significance level.

This does not prove that the system has no effect. It means that there is insufficient statistical evidence to establish a significant reduction in waiting time.